Abhay Bhandari – Gen AI Engineer

SUMMARY

Generative AI Engineer with 3+ years of experience designing and delivering scalable, production-grade AI applications and backend systems. Specialized in **agentic AI**, **multi-agent orchestration**, Retrieval-Augmented Generation (**RAG**), enterprise knowledge search, and LLM-powered automation using **Python**, **FastAPI**, **Azure AI Foundry**, and cloud platforms. Experienced in building secure enterprise integrations, intelligent APIs, and end-to-end AI solutions across sales intelligence, software development, and data analytics. Proven ability to translate complex business requirements into reliable, client-facing products and successfully deploy them in production environments.

SKILLS

- **Core Python & Backend Engineering**
  - Languages & Frameworks: Python, FastAPI, Flask.
  - API & Architecture: REST API Development, Microservices Architecture, API Integration & Debugging.
  - Database Systems: SQL Database Concepts (Relational/Structured).

- **Data Engineering**
  - Pandas, NumPy, Lightweight ETL Pipelines, Data Ingestion (CSV, JSON, APIs, PDFs), Data Cleaning, Validation & Transformation.

- **AI, Machine Learning & Generative AI**
  - GenAI Architectures: Retrieval-Augmented Generation (RAG), Fine-tuning & Custom LLM Pipelines, Prompt Engineering, Agentic AI systems.
  - Orchestration Frameworks: LangChain, LangGraph, Agentic AI, Model Context Protocol (MCP).
  - Vector Databases: Milvus, Chroma, IBM Watson Vector DB.
  - Multi-Agent Systems & LLM Orchestration
  - Azure AI Foundry & Azure AI Search
  - Tool Calling / Function Calling Architectures
  - Supervisor-Agent Design Patterns
  - Real-Time AI Systems

- **DevOps, Cloud & Version Control**
  - Infrastructure: AWS EC2 (Ubuntu/Linux), Docker.
  - Version Control: Git, GitHub.

- **Project Management & Integration Tools**
  - Agile Methodology: Jira (Agile/Scrum).
  - Third-Party Integrations: Intercom, Zendesk Integrations.

Additional LLM based Productivity Tools:

- Developed AI-powered technical assessment and career tools
- Generated interview questions from CVs and automated reports using LLMs
- Built text summarization systems using open-source LLMs
- Implemented OCR-based CV parsing pipelines integrated with LLM workflows

ROLES AND RESPONSIBILITIES

- Designed and developed scalable FastAPI backends for client-facing applications.
- Built and deployed AI-driven RAG and GenAI solutions for enterprise use cases.
- Implemented data preprocessing and transformation pipelines for analytics, ML, and LLM workflows.
- Developed agentic AI systems using LangGraph and MCP for autonomous decision-making.
- Integrated third-party tools such as Intercom and Zendesk for human–AI collaboration.
- Designed agent orchestration frameworks enabling dynamic tool selection and multi-step reasoning workflows
- Deployed microservices on AWS EC2 and managed production environments.
- Collaborated closely with cross-functional teams to deliver solutions aligned with client requirements.
- Actively participated in client demos, debugging sessions, and production issue resolution.

PROJECTS

1. **Jamestown Sales Analytics and Forecasting**

- Analyzed TotalBoat and Amazon sales data to derive actionable sales insights
- Built CI/CD-enabled pipelines for anomaly detection and time-series forecasting
- Reduced model deployment time by ~40%
- Integrated GPT-4 into Slack for on-demand sales intelligence

2. **Pronto CodeLLM – Code Intelligence and QnA System**

- Developed a RAG-based Q&A chatbot for a proprietary programming language.
- Fine-tuned LLMs for domain specific data.
- Implemented FastAPI backend services and integration with Redis for effective prompt/response caching and PostgresSQL for persistent storage.
- Built data preprocessing pipelines and hybrid search (sparse + dense vectors) using Milvus ANN.
- Automated deployments using Docker and managed model execution via Ollama.

3. **Pronto Agentic – Agentic AI for Autonomous Workflow**

- Architected an agentic AI system with MCP support using open-source Ollama models.
- Built FastMCP server for autonomous agent communication.
- Exposed APIs using CopilotKit Python SDK based on client requirements.
- Ensured real-time interaction and secure API integrations.

4. **AI-powered Sales Preparation & Deal Intelligence Platform**

- Built an AI-powered sales preparation platform to generate actionable insights on target companies before client meetings.
- Designed a multi-agent system using Azure AI Foundry to aggregate intelligence from internal data, market signals, and external sources.
- Developed a client intelligence pipeline generating company profiles, therapeutic areas, opportunities, and recent developments.
- Implemented RAG-based internal knowledge retrieval to surface past interactions, projects, and relationship history.
- Integrated external web intelligence and market signals to provide up-to-date

competitive and financial insights.
- Built a decision-making supervisor agent to dynamically select relevant tools and improve response accuracy.
- Integrated MCP-based Salesforce tools for accounts, sales records, opportunities, contracts, contacts, project history, action plans, and competitive intelligence.
- Enabled pre-meeting briefing automation, reducing manual research effort for sales teams.
- Integrated Power BI using an On-Behalf-Of (OBO) authentication flow to securely retrieve and display dashboards tailored to business segments such as Global Consulting Services, Global Product Services, and Worldwide Courier.

5. SDLC Accelerator - Enterprise Knowledge AI and Agentic Delivery Platform

- Built an enterprise knowledge AI search platform that securely ingested organizational content into isolated Weaviate collections, including user guides, test suites, source code, Jira/Azure DevOps issues, and release notes.
- Enabled unified semantic search and an AI-powered knowledge base chatbot across software delivery artifacts while maintaining collection-level separation.
- Developed task-specific agents for story refinement, code-based defect root cause analysis, sprint retrospectives, and engineering metrics reporting.
- Designed a multi-agent architecture with Product Owner, SDK, user-level, and Jira agents to coordinate specialized SDLC workflows and tool interactions.
- Delivered the solution as a production-ready package with admin and client applications, a Django/AJAX frontend, and REST APIs for enterprise integration.
- Implemented data access and authorization logic to securely expose REST endpoints and support integration with external systems.
- Generated BDD-style user stories from Jira data and enabled agent-assisted or manual updates back to Jira.

6. AutoAnalyst - Agentic Data Science and Machine Learning Platform

- Built an agent-powered analytics platform that automated exploratory data analysis, report generation, data preparation, and machine learning workflows from uploaded datasets.
- Enabled autonomous model selection, training, evaluation, and anomaly detection based on the structure and characteristics of the input data.
- Orchestrated specialized agents to execute end-to-end analytical tasks, document training steps, and provide transparent reasoning for key decisions.
- Produced downloadable processed datasets, analytical reports, trained model pickle files, and reproducible training summaries for downstream use.

EXPERIENCE

- **Netsmartz Infotech | Python Developer**
  Sep 2023 – Nov 2025

- **Optimum Data Analytics | Data Scientist**
  Jan 2026 – Present

EDUCATION

- **Bachelor of Technology** (Computer Science & Engineering)